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Technical Section

UniTex: Single-chart texture reconstruction from multi-view images ☆

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ABSTRACT

Reconstructing textures from multi-view images is a fundamental task in computer graphics and computer vision, supporting applications such as virtual production, digital heritage preservation, and content creation. However, existing methods often produce fragmented UV atlases with misaligned seams, which limit their utility in downstream workflows. We present UniTex, a novel framework for reconstructing high-fidelity, single-chart textures from multi-view images. To ensure texture completeness and usability, we introduce a color-aware cut generation algorithm that strategically avoids cuts through salient texture regions, ensuring a seamless, artist-friendly atlas. To improve texture coverage and recover missing details in under-captured regions, we leverage diffusion priors to synthesize novel viewpoints, which provide additional supervision for texture reconstruction. Finally, we employ a physically based differentiable rendering framework that alternately optimizes material and illumination parameters, achieving realistic decoupling of lighting from textures. Extensive experiments demonstrate that UniTex outperforms state-of-the-art methods in visual fidelity, atlas integrity, and practical usability—enabling direct integration into 3D workflows without manual stitching.

1. Introduction

High-quality texture reconstruction from multi-view images is fundamental for applications ranging from virtual production to digital heritage preservation. Early texture reconstruction methods [1–5] primarily focused on alleviating misalignment artifacts caused by geometric errors and camera calibration issues. While these methods can reduce texture tearing and improve surface consistency, their simplified treatment of illumination often leads to baked-in lighting effects, thereby limiting texture reusability in relighting or downstream editing tasks.

With the development of deep learning, several works [6,7] explored disentangling material appearance from lighting by leveraging inverse rendering formulations. More sophisticated models, such as BRDF-based optimization [8] and illumination models grounded in classical rendering [9,10], have further advanced the field by explicitly modeling reflectance. Recent progress in differentiable rendering [11–13] has achieved impressive results in jointly estimating material properties and lighting through physically based light transport simulation. These methods effectively address the decoupling of lighting and material information. However, they primarily focus on the rendering quality, with little attention paid to the texture quality and reusability. As a result, the textures generated by these methods are either fragmented into small pieces (Fig. 2(a)) or of low quality—requiring tedious manual stitching for production use.

To address these challenges, we propose **UniTex**, a unified framework that reconstructs high-fidelity, single-chart textures based on physically based differentiable rendering. The reconstructed textures are production-ready, requiring no post-processing for asset reuse. We first reconstruct the object mesh from the input images. Then, we design a cut generation algorithm that considers the vertex colors of the mesh to produce texture images with improved completeness. During the data collection process, we may encounter cases where the object is insufficiently captured or suffers from self-occlusion. To address this, we utilize diffusion priors [14] to generate novel view images, ensuring sufficient viewpoints for texture optimization. Finally, using a physically based differentiable rendering method, we alternately optimize the lighting and material information to generate textures that are decoupled from lighting (see Fig. 1). In conclusion, our contributions are summarized as follows:

- **High-quality texture.** We propose an efficient framework for recovering high-quality, single-chart textures from multi-view images. By synergistically combining physically based differentiable rendering with diffusion priors, our method effectively decouples illumination while faithfully hallucinating missing details in occluded regions.
- **Cut rationality.** We introduce a more rational cut generation algorithm that incorporates vertex color information, avoiding the

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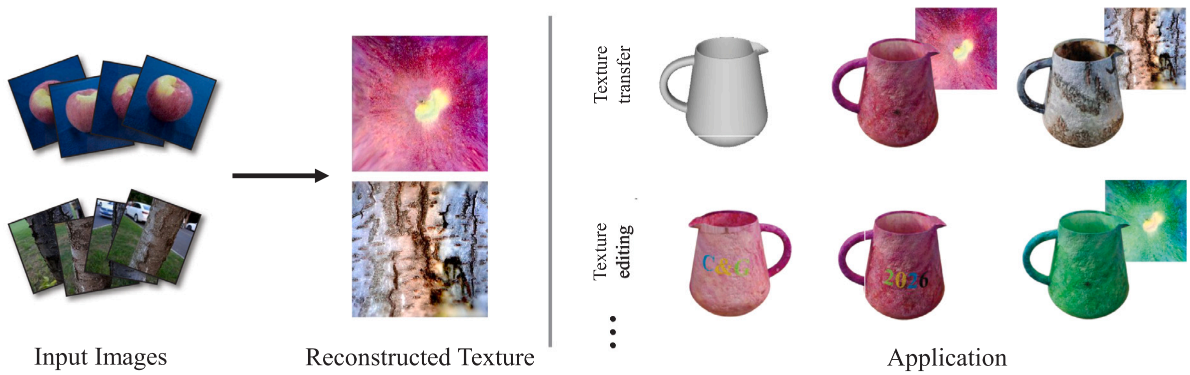


Fig. 1. **UniTex** reconstructs production-ready single-chart textures from multi-view images. The left side shows the reconstructed textures generated by **UniTex**, while the right side presents downstream applications, including texture transfer and texture editing. The inset at the top right of the rendered image displays the corresponding texture map, and the last column in the second row illustrates a recoloring example where the apple texture is modified from red to green.

(a) The result of Fu [1]. (b) Our result.

Fig. 2. (a) Previous methods often overlooked the integrity of the generated texture image, making it unsuitable for direct use in downstream tasks. (b) Our result, with better integrity, is more suitable for downstream tasks.

fragmentation of important texture patterns. This contributes to the generation of more complete and coherent texture maps.

- **Production-ready outputs.** The single-chart textures generated by our method are production-ready. They can be conveniently transferred and applied to other models.

2. Related work

Classical methods. Recovering surface appearance from multi-view images has long been a fundamental topic in computer graphics. Early works such as Lempitsky and Ivanov [15] employ pairwise Markov Random Fields to select the optimal source image for each mesh face. However, these approaches face a major challenge in mitigating visible seams between adjacent face textures. To address this issue, several studies [16,17] introduce additional post-processing procedures, such as multiband blending [18] and Poisson image editing [19], to smooth color transitions across boundaries and reduce visual discontinuities.

To reduce artifacts caused by camera miscalibration and geometric inaccuracy, Zhou and Koltun [5] designed an optimization system, in which both the camera poses and geometric errors are corrected through local image warping. Some methods also adopt a simultaneous optimization of geometry and texture information [3,20,21]. Bi et al. [2] employed a patch-based synthesis strategy to create a new target texture image for each source image, effectively compensating for misalignment resulting from camera drift and reconstructed geometric errors. Despite their progress, these methods rely heavily on post-processing or manual interventions to achieve visually coherent results, and often fail to handle complex surface materials.

Decoupling illumination and material. A major research direction for generating relightable and physically consistent textures is to decouple illumination from material appearance. Early intrinsic decomposition methods aim to separate reflectance and shading, while later neural approaches jointly predict material and lighting parameters [7,8,22,23]. Although effective on controlled datasets, these methods rely on large-scale supervision and lack geometric priors, making them unable to handle interreflections and realistic light attenuation.

In contrast, differentiable rendering provides a physically grounded solution by explicitly modeling light transport. Chen et al. [24] optimize geometry, texture, and illumination via differentiable rasterization, and inverse radiosity [25] extends this to near-field lighting. Subsequent works [11,26,27] support global illumination derivatives for unified material–lighting optimization. Nvdiffrec [28] extends this framework to jointly recover geometry, materials, and environment lighting from multi-view images through physically based differentiable rendering. Hasselgren et al. [29] further improve this by integrating Monte Carlo rendering and denoising for material–light decomposition. However, both methods primarily focus on optimizing per-view appearance and do not include mechanisms for generating complete and coherent texture.

Texture completeness. Generating complete textures with consistent color and detail across the entire surface remains a challenging problem in texture reconstruction. Recent neural methods address this issue by learning continuous appearance fields or constructing unified texture atlases. NeuTex [30] learns a neural texture field conditioned on view direction to produce geometrically aligned texture maps. More recently, Nuvo [31] introduces a neural UV mapping framework to generate consistent and editable full-surface textures from multi-view inputs. However, the learned texture representations in these methods are not physically disentangled from illumination, often resulting in noticeable color discrepancies compared to the actual object surface.

Texture generation and synthesis. Recent advances in diffusion and neural generative models have greatly expanded the capability of automatic texture synthesis. Methods such as TEXTure [32] and Text2Tex [33] employ depth-conditioned diffusion and iterative inpainting to generate textures directly on 3D meshes, but often introduce artifacts at viewpoint junctions. Several works aim to improve spatial and geometric consistency [34–37], while others specialize in human or indoor scene textures [38–41]. In parallel, non-stationary texture modeling explores synthesizing textures with spatially varying statistics, such as the adversarial expansion method [42] and the self-rectification

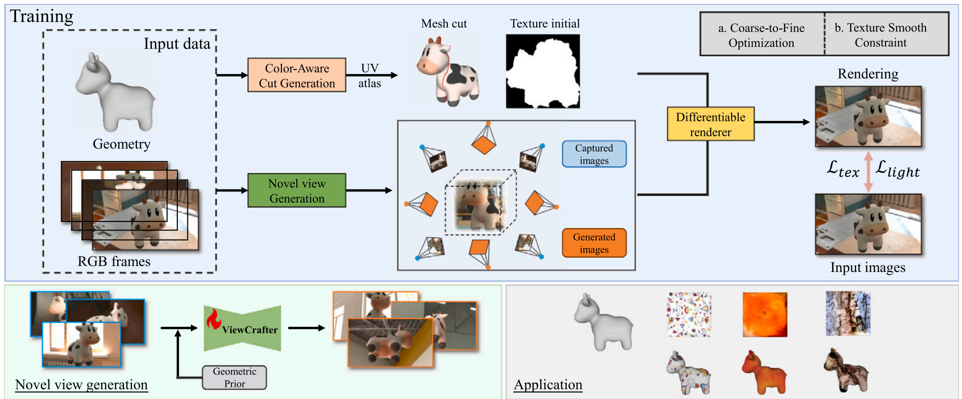


Fig. 3. Pipeline of **UniTex**. The input consists of a sequence of images captured around the object and the reconstructed geometry. To reconstruct single-chart and high-quality texture maps, we create the color-aware mesh cut and generate novel view images by diffusion model prior. To effectively decouple lighting and material information, we adopt an alternating optimization approach. The reconstructed textures can be conveniently transferred and applied to other models.

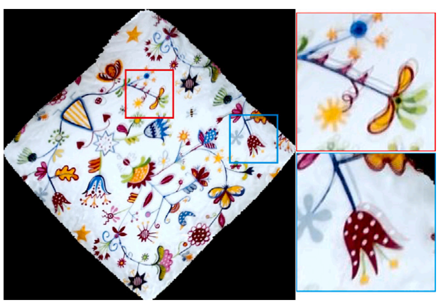


Fig. 4. The blurring on texture obtained without optimizing the camera parameters.

approach [43], which preserve long-range structural variations and non-repetitive patterns. Although these techniques achieve impressive visual diversity, they operate purely in image space without leveraging multi-view or geometric priors, and typically ignore physically based illumination.

3. Method

Method Overview. Our method is designed to reconstruct single-chart textures from multi-view images. To achieve this, we address three key aspects: (1) **Cut generation**. We propose a color-aware cut generation algorithm that strategically avoids cuts through salient texture regions (Section 3.2). (2) **View completion**. To address the challenges of uncaptured regions during the shooting process and self-occlusion of the object, we integrate a diffusion-based model that synthesizes geometrically consistent novel views, thereby enabling texture refinement in missing areas (Section 3.3). (3) **Optimization refinement**. We develop a comprehensive set of optimization strategies, combining physically-based differentiable rendering to generate high-quality textures decoupled from lighting (Section 3.4). Our method pipeline is illustrated in Fig. 3.

3.1. Preprocessing

Key-frames Selection. The input of our method is an RGB sequence or stream acquired by a consumer RGB camera. There is considerable redundant information in the RGB sequence because of the high overlap between consecutive RGB camera scanning views. To reduce the redundant frames and increase computational efficiency, we extract key frames using the method proposed by Fu et al. [3].

Camera Poses Optimization. Due to the drift of the handheld camera trajectory, the generated texture can have blurring and ghosting artifacts, as depicted in Fig. 4. Thus, we optimize the camera parameters using the method proposed by Zhou and Koltun [5].

Mesh Extraction. After extracting keyframes from the RGB sequence, we obtain a collection of multi-view images. We use existing approaches (e.g., NeuS2 [44]) to extract an explicit mesh of the object.

3.2. Single-chart texture reconstruction

Completeness of texture. Creating a complete single-chart texture image is very important for applications because designers can cut out any area as needed. Just as when making clothes, people generally prefer to purchase a whole piece of fabric rather than several separate scraps (as shown in Fig. 2).

Our work focuses on reconstructing high-quality complete textures with topological integrity, ensuring that the texture map represents a **simply-connected** domain and is **homeomorphic to a square**. This topological property facilitates efficient region extraction for downstream texture mapping tasks.

Color-Aware Cutting. To generate the texture image of the object, we need to create a cut for the object mesh. During this process, it is essential that the cut does not pass through salient texture regions. Otherwise, the resulting texture may disrupt important patterns on the object, as illustrated in Fig. 5(a). However, most existing mesh cutting methods rely solely on geometric cues such as curvature, ignoring the intrinsic color distribution of the surface. To address this limitation, we introduce a color-aware formulation that integrates color similarity into the cut construction stage to obtain perceptually consistent cuts.

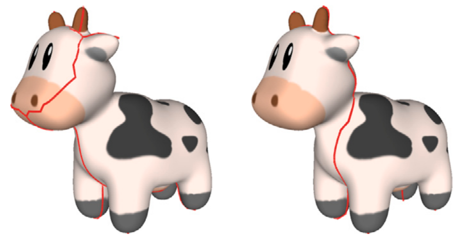
We represent the input mesh as a weighted undirected graph $G = (V, E)$, where each edge $e_{ij} \in E$ connects two adjacent vertices $v_i, v_j \in V$ and is assigned a color-aware weight:

$$w_{ij} = d(c_i, c_j) + \lambda_g \|n_i - n_j\|_2, \tag{1}$$

where c_i and c_j represent the RGB color vectors associated with vertices v_i and v_j, respectively. The term $d(c_i, c_j)$ measures the color discrepancy between these neighboring vertices:

$$d(c_i, c_j) = \sum_{k \in \{r,g,b\}} |k_i - k_j|. \tag{2}$$

Here, k iterates over the red, green, and blue color channels, while k_i and k_j denote the scalar values of the kth color channel for c_i and c_j, respectively. Additionally, $n_i, n_j \in \mathbb{R}^3$ denote the vertex normals at v_i



(a) Without considering vertex color (b) Considering vertex color

Fig. 5. (a) Cut generated without considering *vertex colors*. (b) Cut generated with considering *vertex colors*. Our color-aware cut avoids splitting the salient nose region of the 'spot' model.

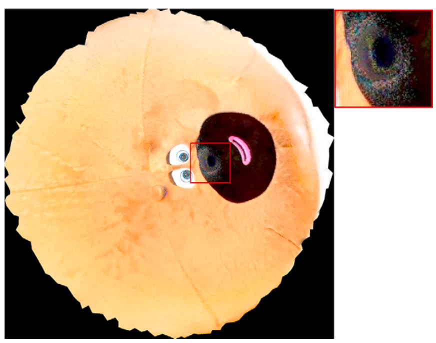


Fig. 6. Insufficient camera viewpoints lead to incomplete supervision in certain regions, resulting in noisy and less reliable texture reconstruction.

and v_j. The geometric term $\|n_i - n_j\|_2$ penalizes large normal deviations, encouraging the cut to remain within geometrically smooth regions. This formulation allows the edge weight w_{ij} to jointly encode both appearance and geometric consistency, so that edges with large color or normal discontinuities are less likely to be selected as part of the final cut.

Building upon the GreedyCut [45] framework, we adopt its point-to-cut strategy to generate the final cut. Specifically, a sparse set of *feature vertices* F is first detected following [45], which marks regions of high geometric distortion or curvature and serves as terminal nodes for cut construction. Given the feature vertices F, the optimal cut is obtained by solving for the minimum-cost tree that connects all terminals:

$$C^* = \arg\min_{C \subseteq E} \sum_{e_{ij} \in C} w_{ij}, \quad \text{s.t.} \quad F \subseteq V(C), \quad C \in \mathcal{T}, \tag{3}$$

where E is the set of mesh edges, w_{ij} is the cost of edge e_{ij} defined in Eq. (1), $V(C)$ denotes the set of vertices covered by C, and $\mathcal{T}$ represents the set of all valid trees in the graph. We employ a *minimum spanning tree* (MST)-based greedy construction to efficiently connect the detected feature vertices. In contrast to the original GreedyCut approach that relies purely on geometric weights, our formulation introduces the color-aware term w_{ij}, enabling the cut to adaptively avoid regions with high texture variation. The resulting cuts maintain smooth geometric transitions and consistent color continuity, as demonstrated in Fig. 5(b). Finally, based on the obtained color-aware cut, we employ the As-Rigid-As-Possible (ARAP) energy [46] for mesh flattening to minimize parameterization distortion.

3.3. Texture completion with diffusion prior

Insufficient camera viewpoints and self-occlusion during the capture process cause incomplete color constraints in texture optimization. This leads to noise artifacts (Fig. 6) and blank regions, which in turn degrade the quality and completeness of the texture. To address this issue, we leverage recent advances in diffusion-based novel view synthesis techniques, applying the ViewCrafter [14] model to generate novel views for unseen viewpoints. These synthesized views are combined with the original captures to provide dense and diverse supervision for texture refinement.

While ViewCrafter [14] effectively improves view coverage, its point-cloud-based optimization may suffer from spatial drift and boundary inconsistency. To further enhance its geometric reliability, our method incorporates a reconstructed mesh as input, providing the diffusion model with an explicit and structured geometric prior.

To explicitly encode geometric information within the diffusion model, we render three geometry maps from the mesh (depth, normal map, and visibility mask) and normalize them into a multi-channel tensor:

$$\mathbf{C}_{\text{geom}} = [D, N_x, N_y, N_z, M], \tag{4}$$

where D denotes the per-pixel depth, (N_x, N_y, N_z) the surface normal components in the camera coordinate frame, and M represents the binary visibility mask. This geometry tensor $\mathbf{C}_{\text{geom}} \in \mathbb{R}^{H\times W\times 5}$ is passed through a lightweight convolutional encoder $\mathcal{A}(\cdot)$, referred to as the *Adapter*, which maps it to a latent-space feature:

$$\hat{\mathbf{Z}}_{\text{cond}} = \mathcal{A}(\mathbf{C}_{\text{geom}}). \tag{5}$$

The encoded feature is concatenated with the noisy latent variable $\mathbf{Z}_t$ in the diffusion process, and then passed through a 1×1 convolution to align the fused representation with the channel dimension of the diffusion model's latent space:

$$\mathbf{Z}'_t = \text{Conv}_{1\times1}\big([\mathbf{Z}_t \mid \hat{\mathbf{Z}}_{\text{cond}}]\big), \tag{6}$$

where $\mathbf{Z}'_t$ denotes the geometry-conditioned latent feature. The fused latent is then fed into the diffusion U-Net for denoising, following the standard training objective:

$$\mathcal{L}_{diff} = \mathbb{E}_{\mathbf{Z}_0,\epsilon,t}\big[\|\epsilon - \epsilon_\theta(\mathbf{Z}'_t, t)\|^2\big], \tag{7}$$

where ϵ_θ is the predicted noise and t denotes the diffusion timestep. Conditioned jointly on appearance and geometry, the diffusion model synthesizes novel views that remain consistent with surface structure and visibility across viewpoints. Overall, our method combines the generative capability of ViewCrafter with explicit mesh priors to achieve geometrically consistent novel-view synthesis, providing stable and controllable supervision that enhances texture completeness and cross-view consistency.

3.4. Optimization framework

To reconstruct high-quality textures from multi-view images, it is crucial to correctly model the interaction between illumination and surface reflectance, as lighting variations across viewing directions can significantly alter the perceived appearance of the material. While recent neural rendering methods (e.g., NeuTex [30] and Nuvo [31]) attempt to recover textures directly by learning view-dependent appearance through MLPs, these approaches often lack an explicit illumination model. Consequently, the optimized textures may deviate from the true diffuse appearance, as illustrated in Fig. 7. To achieve physically consistent texture recovery, we adopt a physically-based differentiable rendering framework for joint optimization of material and illumination.

Physically-based optimization The rendering equation models light interaction with surfaces, accounting for their material properties and geometry. It is given by:

$$L_o(\mathbf{x}, \omega_o) = \int_\Omega L_i(\mathbf{x}, \omega_i) f(\mathbf{x}, \omega_i, \omega_o)(\omega_i \cdot \mathbf{n})\, d\omega_i, \tag{8}$$

Fig. 7. Limited texture quality of neural rendering methods. (a) Input multi-view images. (b) Texture reconstructed by NeuTex [30]. Due to the absence of explicit physically-based rendering, NeuTex (likewise Nuvo [31]) fails to decouple illumination from the surface appearance, resulting in significant color discrepancies and visual artifacts.

where $\mathbf{x}$ and $\mathbf{n}$ are the surface point and its normal vector. f models the BRDF properties, and L_i and L_o denote the incoming and outgoing light radiance in directions ω_i and ω_o. Ω signifies the hemispherical domain above the surface. Over the past few decades, various general and specialized BRDF models have been proposed [47]. The Disney BRDF [48] is a widely adopted material model in films and games. It features ten high-level parameters, such as base RGB color, metallicity, roughness, and clearcoat.

However, in unstructured optimization, such a high-dimensional parameter space often leads to ambiguity, where multiple parameter configurations can produce visually similar appearances. To reduce this ambiguity and stabilize optimization, we constrain our formulation to handle only opaque surfaces with three essential parameters: **diffuse reflectance**, **roughness**, and **specular intensity**. This compact yet expressive representation captures the dominant appearance variations while maintaining physical interpretability and robustness during optimization. Illumination is modeled using an environment light map that approximates complex global lighting conditions. During optimization, both material and illumination parameters are jointly refined through differentiable rendering, ensuring a physically consistent separation of lighting and texture. This physically grounded design enables our method to reconstruct faithful diffuse textures that remain consistent under novel lighting conditions.

Training Strategy. Due to the highly ill-posed nature of the inverse rendering problem described above, starting the optimization with all unknowns of a large scene at their highest resolution leads to low-quality local optima. We design a coarse-to-fine optimization strategy that gradually increases the dimensionality of the variables. For the diffuse texture and environment map lighting, we first optimize at a low resolution and then progressively increase the resolution. Furthermore, to better decouple the lighting and material information, we employ an alternating optimization approach, iteratively optimizing the lighting information and the material information.

Loss functions design. We expect that the BRDF properties do not change drastically in areas with smooth color. We define a smooth constraint as:

$$\mathcal{L}_{smooth} = \|D(x) - D(x+\epsilon)\|_2, \tag{9}$$

where D denotes the texture maps, x is the pixel position, and ϵ is a random 2D displacement vector.

In addition, we use pixel-wise mean squared loss $\mathcal{L}_{mse}$ and light regularization loss $\mathcal{L}_{reg}$ assuming a near-natural white incident light:

$$\mathcal{L}_{mse} = \mathrm{MSE}(C(\theta),\ C_{gt}), \tag{10}$$

$$\mathcal{L}_{reg} = \sum_c (L_c - \frac{1}{3}\sum_{c'} L_{c'})^2,\ c, c' \in \{R, G, B\}, \tag{11}$$

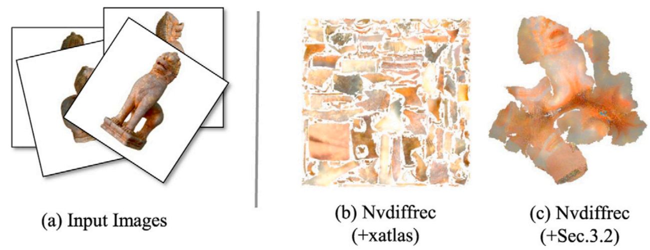


Fig. 8. Comparison of texture reconstruction. Baseline methods produce fragmented texture maps (b) that are difficult to reuse. In (c), we replace the xatlas parameterization in Nvdiffrec [28] with our method (Section 3.2).

where $C(\theta)$ and C_{gt} represent the rendered image and the ground truth, respectively. θ represents the parameters in the scene, including lighting information, material properties, etc. L_c is the cth color channel of the environment light map.

In our optimization process, we alternately fix the lighting and optimize the material information, then fix the material and optimize the lighting information. In the material optimization stage, we use the following loss function:

$$\mathcal{L}_{tex} = \mathcal{L}_{mse} + \lambda_1 \mathcal{L}_{smooth}. \tag{12}$$

Meanwhile, during the lighting optimization stage, we utilize the following loss function:

$$\mathcal{L}_{light} = \mathcal{L}_{mse} + \lambda_2 \mathcal{L}_{reg}, \tag{13}$$

where λ_1 and λ_2 are hyperparameters. The alternating optimization between material and lighting parameters converges reliably in practice. Both subproblems minimize strongly coupled energy functions, $\mathcal{L}_{tex}$ and $\mathcal{L}_{light}$, which share the photometric reconstruction term $\mathcal{L}_{mse}$. Empirically, the $\mathcal{L}_{mse}$ term dominates the total objective by one to two orders of magnitude, leading to a consistently decreasing global energy across alternating iterations.

4. Experiments

4.1. Experimental settings

Dataset and Metrics. We evaluate our method on three datasets with diverse styles to verify its effectiveness. The real-world dataset contains seven objects captured under varying illumination and material conditions using a Nikon Z30 camera. The synthetic dataset includes two objects from Nvdiffrec [28]. We additionally test on the ‘lion’ model reconstructed from a 3D scan of the *Bayon Lion* statue at the Musée Guimet in Paris, demonstrating the ability of our approach to handle geometrically complex models.

We report metrics including the peak signal-to-noise ratio (PSNR), structural similarity (SSIM [49]), and perceptual similarity as measured by LPIPS [50] for the rendering results. Due to the absence of ground-truth texture references for real-world objects, we employ a total variation (TV) regularization term [51] to quantitatively assess the smoothness and stability of reconstructed textures. This metric penalizes local intensity fluctuations and serves as an indicator of spatial consistency across the generated textures. For synthetic data with ground-truth material attributes, we additionally quantify texture reconstruction fidelity using the ‘PSNR (texture)’ metric computed on the synthesized textures. While our method can optimize metallicity, roughness, and diffuse texture maps, we restrict texture quality evaluation to diffuse channels due to their superior perceptual objectivity and greater practical significance in real-world applications. In our experiments, we use 10% of the captured keyframes as the test set

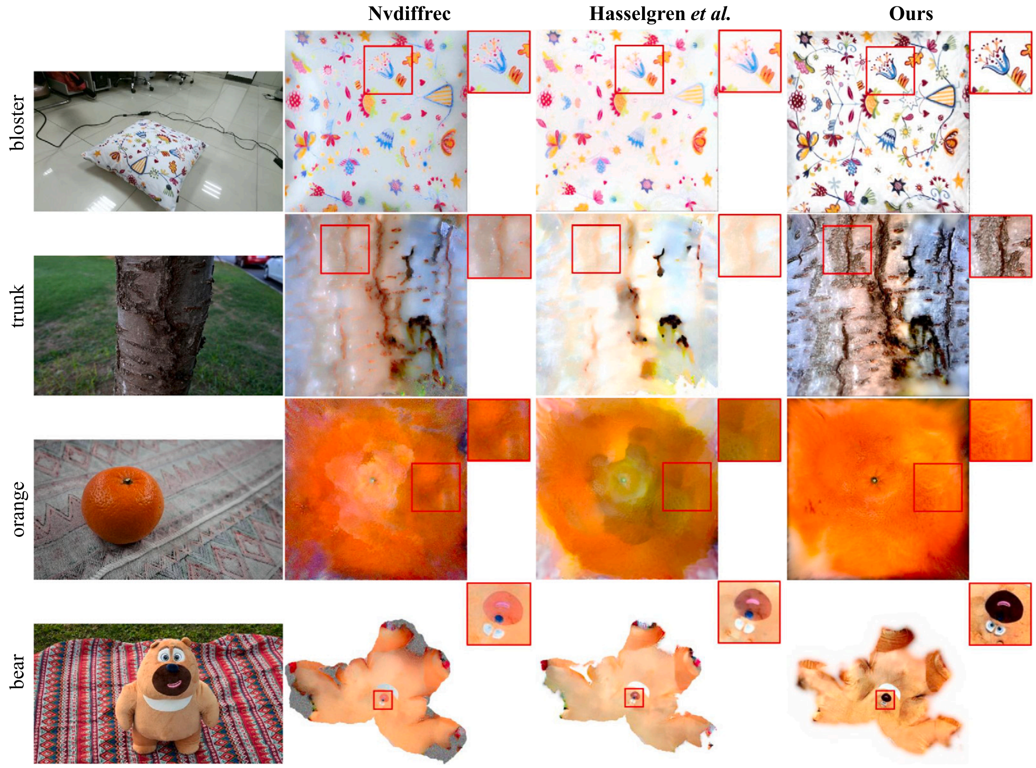


Fig. 9. Comparison of texture reconstruction results. The first column shows single-view inputs of different models, while the remaining columns present textures reconstructed by state-of-the-art methods and our approach. Insets at the upper-right corners highlight detail differences, demonstrating the superior quality of our generated textures.

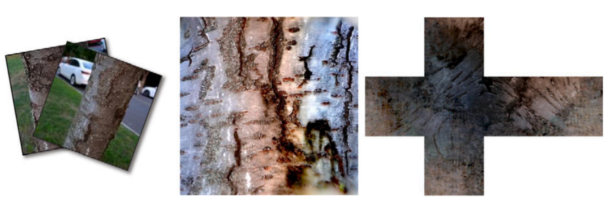


Fig. 10. Qualitative comparison against NeuTex. (a) Reference images. (b) Our result. (c) Texture reconstructed by NeuTex. Compared to NeuTex, our method exhibits noticeably less parametric distortion and recovers more faithful intrinsic colors.

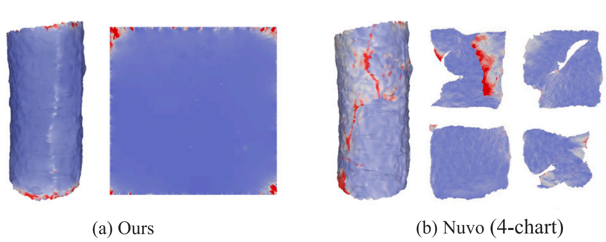


Fig. 11. Qualitative comparison of UV mapping distortion against Nuvo, visualizing distortion magnitude from blue (low) to red (high). While Nuvo exhibits noticeable local distortion, our method achieves a significantly lower-distortion parameterization.

and the remaining 90% for training to objectively assess the rendering quality of our model (see Fig. 12).

Compared Baselines. Although extensive research has been conducted on material and lighting decomposition, including methods based on inverse rendering [12] and novel view synthesis [52], these methods primarily focus on achieving high-fidelity rendering results and do not pay much attention to generating complete texture images. To comprehensively evaluate our method, we select highly relevant state-of-the-art (SOTA) baselines: Nvdiffrec [28], Hasselgren et al. [29], NeuTex [30], and Nuvo [31]. Specifically, Nuvo is a neural UV mapping approach built upon the NeuTex framework. Since its official source code is not publicly available, we carefully re-implemented this method based on the original paper. While NeuTex and Nuvo explore texture completeness, they typically recover textures by learning view-dependent appearance via MLPs. Consequently, these approaches often lack an explicit illumination model and struggle to decouple material properties from lighting, as evidenced by the artifacts in Fig. 7(b). Nvdiffrec and Hasselgren et al. explicitly generate object materials from multi-view images via differentiable rendering. To ensure a fair comparison in terms of texture quality, we use the same parameterization method for each object during optimization (see Fig. 8(c)).

Implementation Details. To reduce redundancy and improve computational efficiency, we extract key-frames following Fu et al. [3]. A new frame i is selected and added to the key-frame set Φ_{KF} if its camera pose satisfies:

$$C_i = \{C_i \in \Phi_{KF} : \angle(\mathrm{R}_k, \mathrm{R}_i) > 30^\circ \quad \| \quad \mathrm{Dist}(\mathrm{t}_k, \mathrm{t}_i) > 0.2\}, \tag{14}$$

where k is the index of the previously selected key-frame. $\angle(\mathrm{R}_k, \mathrm{R}_i)$ and $\mathrm{Dist}(\mathrm{t}_k, \mathrm{t}_i)$ denote the angle between their rotation matrices and the Euclidean distance between their translation vectors, respectively. These rotational (30°) and translational (0.2 m) thresholds are empirically set based on their configuration. Based on the extracted Φ_{KF}, we dynamically determine novel viewpoints to ensure comprehensive omnidirectional coverage. We sample a predefined total of N_{total} uniform poses (empirically set to 50) on a bounding sphere fitted to the input camera trajectories using a spherical Fibonacci point set [53], and replace the ones closest to the key-frames with these real captures. The remaining unreplaced viewpoints determine the exact poses for novel

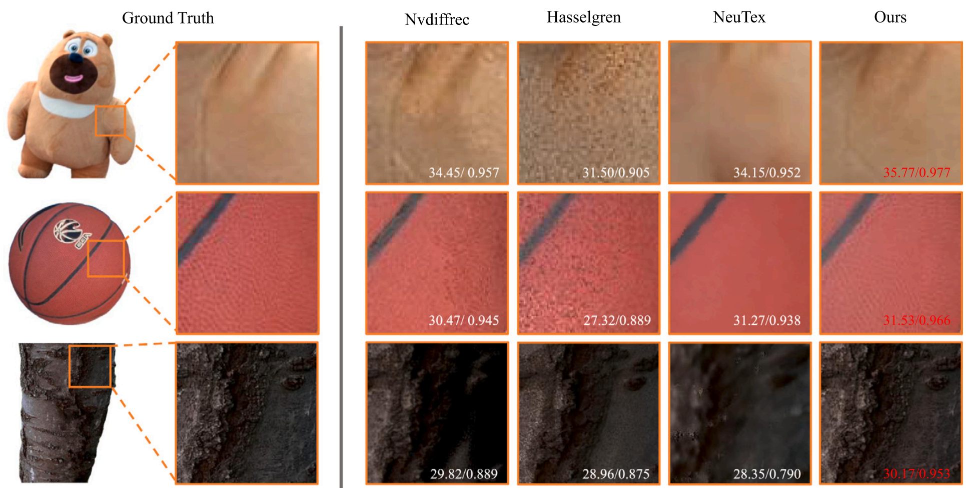


Fig. 12. Qualitative comparison of rendered results on the real-world dataset (close-up views). PSNR and SSIM values are shown below each rendering.

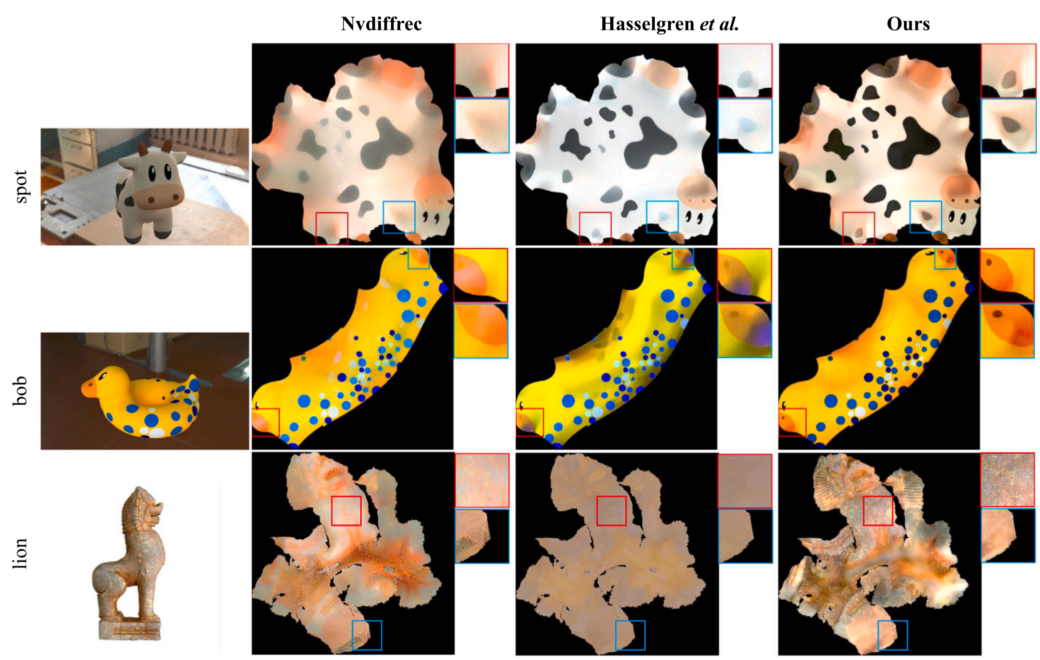


Fig. 13. Comparison of texture reconstruction results on the 'spot', 'bob', and 'lion' models. Insets at the upper-right corners highlight regions where our method achieves higher reconstruction fidelity.

view synthesis, allowing the generation count to naturally adapt to the sparsity of the input captures. The optimization process of our method is implemented using Mitsuba3 [54]. We empirically set $\lambda_g = 10.0$, $\lambda_1 = 1 \times 10^{-2}$ and $\lambda_2 = 2.0$ (in Eq. (1), Eqs. (12) and (13)). In the alternating optimization stage, we alternate between material and lighting updates, performing 30 iterations for material optimization followed by 20 iterations for lighting optimization in each cycle. The texture resolution is set to 2048 × 2048. We employ the As-Rigid-As-Possible (ARAP) [46] parameterization method for models. For objects with simple geometric structures (such as 'apple', 'basketball', 'bloster', and 'orange'), we implement fixed-boundary parameterization to ensure texture map regularity, thereby enhancing compatibility with downstream pipelines. In the diffusion module, we adopt a lightweight design that introduces only a small number of trainable parameters to align the geometric conditioning within the latent space. The Adapter encoder module in Eq. (5) processes the 5-channel geometric tensor through three consecutive 3 × 3 convolutional layers with 16, 32, and 64 output channels, respectively. After concatenating this 64-channel geometric feature with the 4-channel noisy latent variable Z_t, a 1 × 1 convolution in Eq. (6) projects the combined 68 channels back to 4 channels. This ensures the output feature remains strictly aligned with the latent representation of the diffusion U-Net. During training, we fine-tune the spatial attention layers of the ViewCrafter U-Net using a learning rate of 1×10^{-5} and a batch size of 4, while keeping the VAE encoder and decoder frozen. All experiments are conducted on a single NVIDIA RTX A6000 GPU. The diffusion module training requires approximately 2 h, while the subsequent texture and lighting optimization takes about 30 min for each scene.

Table 1
Quantitative comparison of rendering quality and texture TV loss on the real-world dataset. The **best** and second-best results are highlighted.

Scene	Method	PSNR ↑	SSIM ↑	LPIPS ↓	TV ↓
apple	Nvdiffrec [28]	30.02	0.953	0.061	0.0910
	Hasselgren et al. [29]	29.87	0.915	0.078	0.0401
	NeuTex [30]	29.15	0.921	0.079	0.0619
	Ours	**30.96**	**0.964**	**0.040**	**0.0298**
bear	Nvdiffrec [28]	32.27	0.960	0.059	0.0320
	Hasselgren et al. [29]	28.23	0.902	0.088	0.0480
	NeuTex [30]	32.18	0.951	0.067	0.0539
	Ours	**33.18**	**0.973**	**0.037**	**0.0240**
Doraemon	Nvdiffrec [28]	**29.86**	0.964	0.052	0.0415
	Hasselgren et al. [29]	28.39	0.933	0.068	**0.0046**
	NeuTex [30]	28.65	0.950	0.067	0.0113
	Ours	28.73	**0.966**	**0.041**	0.0081
trunk	Nvdiffrec [28]	28.13	0.944	0.061	0.0992
	Hasselgren et al. [29]	27.23	0.933	0.067	0.0067
	NeuTex [30]	27.99	0.941	0.059	0.0266
	Ours	**28.47**	**0.948**	**0.051**	**0.0030**
orange	Nvdiffrec [28]	**25.37**	**0.916**	**0.099**	0.0233
	Hasselgren et al. [29]	24.14	0.832	0.147	0.0206
	NeuTex [30]	24.31	0.887	0.127	0.0318
	Ours	24.88	0.841	0.124	**0.0133**

Table 2
Comparison of texture extraction time and computational overhead against baseline methods.

Method	Texture extraction	Peak VRAM
Nvdiffrec [28]	41.6 mins	8.3 GB
Hasselgren et al. [29]	38.5 mins	10.6 GB
NeuTex [30]	397.5 mins	7.2 GB
Ours	**35.6 mins**	**4.3 GB**

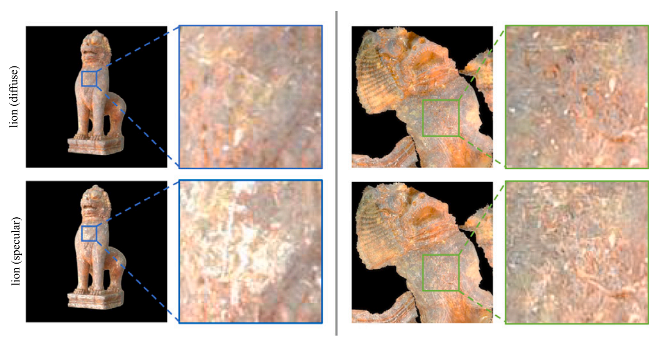


Fig. 14. Results on the 'lion' model under diffuse and specular materials. Left: rendered appearances. Right: corresponding reconstructed textures. As shown in the zoomed-in views, our method successfully suppresses specular highlights to recover accurate diffuse textures.

4.2. Results analysis

We validate the effectiveness of our method through comprehensive experiments on texture reconstruction quality, rendering quality, and performance on complex materials.

Qualitative Analysis. We qualitatively analyze our method in terms of texture quality, rendering quality, and its ability to handle complex materials while accurately decoupling illumination.

As shown in Figs. 9 and 13, our method produces textures with significantly higher fidelity compared to existing methods. Nvdiffrec [28] introduces noticeable noise artifacts along texture boundaries (see the 'bear' in Fig. 9). For Hasselgren et al. [29], the reconstructed textures exhibit a clear loss of high-frequency details (as observed in the 'bloster' and 'trunk' models in Fig. 9, and the 'lion' model in Fig. 13). In addition, the results of Hasselgren et al. show material–lighting ambiguity, leading to visible artifacts such as the distorted texture on the 'orange' model. In contrast, our optimization framework more accurately decouples the influence of illumination from the intrinsic texture appearance, enabling the reconstruction of diffuse textures that better match the true surface characteristics of the objects. Figs. 10 and 11 qualitatively compare our method against two neural baselines: NeuTex, a neural rendering approach, and Nuvo, a neural UV mapping technique. Lacking explicit geometric regularization and a physically-based rendering formulation, NeuTex suffers from severe parametric distortion and fails to fully decouple illumination, yielding color discrepancies in the extracted textures. While Nuvo builds upon NeuTex with a point-sampled discrete optimization for UV mapping, this formulation cannot guarantee bijectivity. Its reliance on multiple competing soft penalty losses increases susceptibility to local minima, frequently causing folded or flipped triangles. Consequently, as visualized in Fig. 11 using the symmetric Dirichlet energy [55] to quantify isometric stretching, Nuvo produces parameterizations with noticeably higher geometric distortion than our approach. Additionally, Fig. 12 illustrates the qualitative advantages of our approach in producing highly realistic and illumination-consistent renderings.

Table 3
Quantitative comparison of textures on the synthetic dataset. The **best** and second-best results are highlighted.

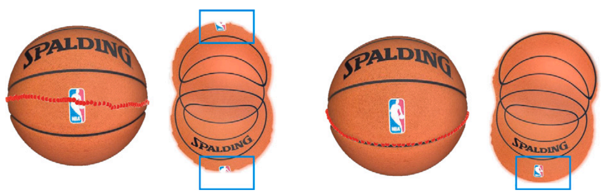
Scene	Method	PSNR (texture) ↑	TV ↓
spot	Nvdiffrec [28]	31.65	0.0057
	Hasselgren et al. [29]	28.24	0.0051
	NeuTex [30]	29.53	0.0060
	Ours	**33.45**	**0.0046**
bob	Nvdiffrec [28]	31.50	0.0054
	Hasselgren et al. [29]	29.68	**0.0035**
	NeuTex [30]	30.12	0.0050
	Ours	**32.48**	0.0049

To further verify that our method can effectively handle complex materials and accurately decouple illumination, we conduct an additional experiment on the 'lion' model rendered under two different material settings: 'lion (diffuse)' and 'lion (specular)'. The 'lion (specular)' model exhibits strong reflective highlights, as illustrated in the close-up renderings in Fig. 14. Even under this challenging specular condition, our method successfully separates lighting effects from the underlying texture, producing faithful diffuse reconstructions. As shown by the extracted textures, both 'lion (diffuse)' and 'lion (specular)' achieve high fidelity and remain consistent with the object's true surface appearance, demonstrating the robustness of our approach in disentangling illumination from material properties.

Quantitative Analysis. To further validate the accuracy of texture reconstruction, we quantitatively evaluate the synthetic dataset with ground-truth textures using two metrics: PSNR (texture) and Total Variation (TV) loss. As reported in Table 3, our method achieves higher PSNR and lower TV loss values than competing approaches, indicating closer alignment with ground-truth diffuse textures and more effective suppression of high-frequency noise. To further assess rendering quality, Table 1 compares PSNR, SSIM, and LPIPS metrics computed on rendered images. Our method generally attains higher scores across most metrics, demonstrating that the reconstructed materials and illumination parameters yield photometrically accurate and perceptually consistent renderings. Moreover, the lower TV loss values reported in Table 1 further indicate that our reconstructed textures maintain high quality even in real-world scenes, with smoother shading transitions and fewer noise artifacts. These quantitative results collectively confirm that our physically based optimization framework improves texture fidelity and enhances rendering realism under varying lighting conditions. Furthermore, we evaluate the optimization efficiency of our pipeline. As reported in Table 2, our texture extraction runtime is highly competitive with explicit differentiable rendering baselines, such as Nvdiffrec and Hasselgren et al. and is significantly faster than neural rendering approaches like NeuTex.

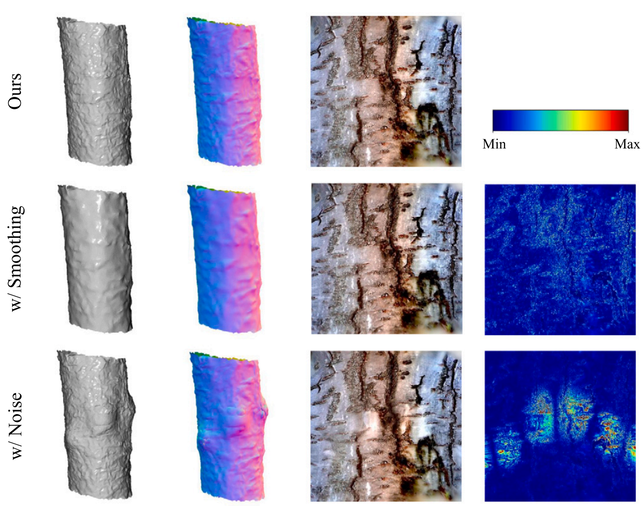
Table 4
Parameter sensitivity analysis on the trade-off between detail preservation (PSNR, SSIM) and texture smoothness (TV loss). Our configuration (×1.0) optimally balances the loss magnitudes, preventing over-smoothing while maintaining high rendering fidelity.

Parameter Setup		PSNR↑	SSIM↑	TV↓
Optimization Weight (λ_1)	×0.1	29.42	0.940	0.0285
	×0.5	29.51	0.942	0.0195
	×2.0	29.35	0.935	0.0125
	×10.0	28.10	0.912	**0.0098**
Optimization Weight (λ_2)	×0.1	28.49	0.925	0.0212
	×0.5	29.15	0.931	0.0188
	×2.0	29.20	0.937	0.0165
	×10.0	28.45	0.920	0.0173
Geometric Weight (λ_g)	×0.1	29.54	0.938	0.0188
	×0.5	29.55	0.940	0.0181
	×2.0	29.54	0.942	0.0186
	×10.0	29.53	0.941	0.0192
Our configuration (×1.0)		**29.56**	**0.943**	0.0156



(a) Without Color-Aware Cutting (b) Ours

Fig. 15. Ablation on color-aware cut generation. Red path indicates the generated cut. (a) Without color cues, cuts arbitrarily cross salient features, introducing visual discontinuities. (b) Our method adaptively routes cuts around high-variation regions, preserving texture integrity.



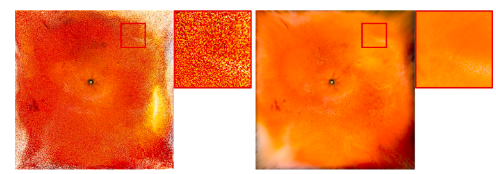
(a) Mesh (b) Mesh Normal (c) Texture (d) Texture Deviation

Fig. 16. Impact of geometric inaccuracies on texture reconstruction.

Table 5
Ablation study on real-world scenes. We evaluate variants omitting color-aware cut generation (w.o. color-aware), diffusion priors (w.o. DP), the smoothness loss (w.o. $\mathcal{L}_{smooth}$), and alternating optimization (w.o. AO). Evaluation is based on PSNR, SSIM, and LPIPS for novel view renderings, and TV loss for reconstructed textures on real-world scenes.

Methods	PSNR↑	SSIM↑	LPIPS↓	TV↓
w.o. color-aware	29.52	0.941	0.051	0.0265
w.o. DP	27.15	0.898	0.088	0.0257
w.o. $\mathcal{L}_{smooth}$	28.49	0.926	0.059	0.0312
w.o. AO	28.37	0.934	0.057	0.0289
Full model	**29.56**	**0.943**	**0.048**	**0.0156**

(a) Comparison without (left) and with (right) diffusion priors.



(b) Comparison without (left) and with (right) the smoothness loss $\mathcal{L}_{smooth}$.

Fig. 17. Ablation study on (a) diffusion priors and (b) the smoothness loss $\mathcal{L}_{smooth}$.

4.3. Ablation study

To validate the individual contributions of our proposed components and algorithmic designs, we perform comprehensive ablation studies through both qualitative and quantitative evaluations.

As shown in Fig. 5, the color-aware cutting module effectively prevents cuts from passing through salient texture regions, avoiding disruption of important surface patterns. To explicitly validate this design, Fig. 15 presents a qualitative ablation on the color-aware cut generation. As shown in Fig. 15(a), without color constraints, the cut arbitrarily passes directly through salient texture features, leading to noticeable structural discontinuities along chart boundaries. By contrast, our method routes the cut around these high-frequency regions (Fig. 15(b)), effectively preserving the visual integrity of the extracted texture. Fig. 17 presents qualitative comparisons on the diffusion priors and the smoothness loss $\mathcal{L}_{smooth}$. Incorporating diffusion priors significantly enhances texture completeness and fine-scale detail, while the smoothness loss $\mathcal{L}_{smooth}$ effectively suppresses high-frequency noise, resulting in more refined and visually consistent texture maps.

Furthermore, we evaluate our robustness against geometric inaccuracies by applying explicit smoothing and vertex noise to the initial reconstructed meshes (Fig. 16). Minor geometric deviations (w/ Smoothing) have a negligible impact on texture quality. Severe geometric perturbations (w/ Noise) inevitably introduce parametric distortion. In practice, the adopted object-centric reconstruction of NeuS2 provides highly reliable geometric initializations across our tested datasets. The inherent geometric errors remain within a constrained range, thereby exerting minimal impact on the synthesized textures. For challenging scenarios where accurate geometric reconstruction remains difficult, extending the current differentiable rendering framework to support joint optimization of geometry and texture can mitigate these geometric errors, presenting a rigorous direction for future research.

Table 4 presents our parameter sensitivity analysis, where we scale our weights ($\lambda_1, \lambda_2, \lambda_g$) by factors of $\{0.1, 0.5, 2.0, 10.0\}$. The weight λ_1 balances texture smoothness and detail preservation. A large λ_1 (×10.0) overly smooths the texture, reducing TV loss (0.0098) but degrading rendering PSNR (28.10), whereas a small λ_1 (×0.1) insufficiently penalizes noise. The weight λ_2 regularizes the optimized illumination to approximate natural white light. Deviating from the optimal setting (×0.1 or ×10.0) violates this natural lighting prior, subsequently deteriorating both texture quality and rendering fidelity (PSNR drops below 28.50). Finally, the geometric weight λ_g controls the cut generation.

Fig. 18. Applications of our method. Reconstructed production-ready textures are transferred onto different models to demonstrate reusability. The last column shows a color-editing result, where the apple texture is recolored from red to green.

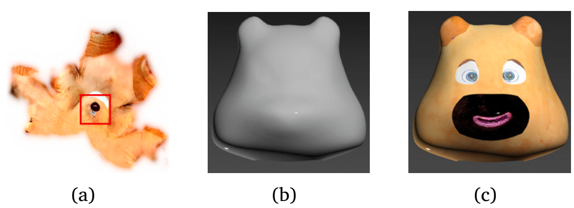


Fig. 19. A portion of the 'bear' texture (a) is extracted and transferred onto a newly created animal model (b). The resulting textured model is shown in (c).

While it directly affects the texture layout and smoothness (TV loss), its impact on the final rendering quality is negligible (PSNR remains stable around 29.54). Overall, our proposed configuration (×1.0) achieves the optimal trade-off between high-fidelity rendering and stable texture generation.

We validate the effectiveness of our core algorithmic components, with quantitative results summarized in Table 5. The variant without color-aware cut generation is denoted as *w.o. color-aware*, the version without diffusion priors as *w.o. DP*, the version without the smoothness loss $\mathcal{L}_{smooth}$ as *w.o.* $\mathcal{L}_{smooth}$, and the version without the alternating optimization strategy for material-lighting decomposition as *w.o. AO*. As shown in the table, while the *w.o. color-aware* variant maintains comparable novel view rendering metrics (e.g., PSNR and SSIM) to the full model, it yields a higher Total Variation (TV) loss. This increase in TV loss occurs because, without color constraints, the cut can pass directly through salient texture features, introducing noticeable structural discontinuities along chart boundaries. In addition, removing diffusion priors leads to a substantial drop in rendering quality, demonstrating their importance for recovering fine texture details from limited viewpoints. Furthermore, omitting either $\mathcal{L}_{smooth}$ or the alternating optimization scheme results in approximately a 1.0 dB decrease in PSNR compared with the full model. Our complete framework also achieves the lowest TV loss among all variants, indicating smoother and more visually stable reconstructed textures.

4.4. Applications and discussion

We demonstrate the utility of our reconstructed textures through texture transfer and color editing. As shown in Fig. 18, the reconstructed textures ('bolster', 'apple', and 'trunk') are easily mapped onto different meshes, demonstrating their reusability. The rightmost example further shows editability by recoloring the apple texture. Additionally, Fig. 19 illustrates transferring a portion of the bear texture onto a novel animal head model. Collectively, these applications

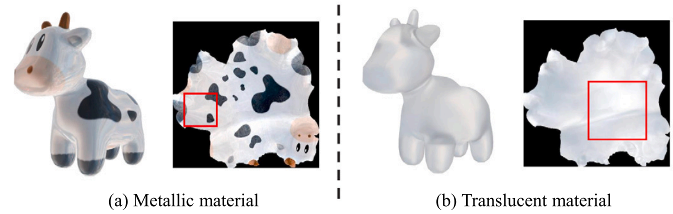


Fig. 20. Limitations on challenging non-Lambertian surfaces. Highly metallic or translucent materials may still exhibit residual baked-in illumination due to complex light transport.

indicate that our method produces production-ready textures integrable into standard 3D creation pipelines.

While effective across typical assets, recovering the intrinsic properties of extreme non-Lambertian materials remains challenging. As illustrated in Fig. 20, highly metallic surfaces with complex view-dependent reflections and translucent objects governed by subsurface scattering can occasionally exhibit residual baked-in artifacts.

Furthermore, mapping complex 3D topologies onto a single UV chart inherently introduces geometric distortion. Our framework mitigates this by combining *GreedyCut*, which routes seams through local maximizers of isometric distortion, with ARAP flattening to distribute residual stretching. This mechanism maintains the "single-chart" property while preventing severe texture shearing. For extreme topologies, our pipeline accommodates advanced singularity-driven parameterization techniques [56,57]. Integrating these methods to absorb intrinsic Gaussian curvature via sparse cone singularities would further alleviate texture stretching.

5. Conclusion and future work

We presented UniTex, a novel framework for reconstructing high-quality single-chart textures from multi-view images. Our method addresses the critical limitations of prior work: fragmented atlases, baked lighting artifacts, and incomplete texture coverage. Key advantages include: (1) Production-ready outputs. Unified texture charts eliminate manual stitching, enabling direct use in 3D workflows. (2) Lighting robustness. Our optimization pipeline reliably decouples diffuse albedo from illumination, even in complex real-world captures. (3) Perceptual quality. Color-guided cutting and diffusion priors preserve structural patterns and fill occlusions plausibly.

While UniTex effectively recovers intrinsic textures for opaque surfaces and standard specularities, extreme non-Lambertian materials, such as highly metallic or translucent objects, remain a challenge. Extending our framework to model advanced material properties (e.g., layered BSDFs) and exploring real-time adaptation for dynamic scenes represent key directions for future work. UniTex demonstrates that bridging classical parameterization with modern inverse rendering resolves long-standing texture fragmentation problems, opening new possibilities for asset generation in graphics and vision.

CRediT authorship contribution statement

Husen Li: Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dong Xiao:** Writing – review & editing, Supervision. **Jinghao Zhang:** Visualization, Validation, Software, Data curation. **Renjie Chen:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2026.104599.

Data availability

Code and data will be made publicly available upon acceptance of the article.

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